

GymAkhada Project Files and Folder Guide

Yeh PDF project ke har important folder/file ka kaam simple Hinglish me explain karta hai. Isme yeh bhi bataya gaya hai ki kis file me kaisa code likha hai aur runtime par woh kya kaam karta hai.

Project path	D:/GymAkhada
Framework	ASP.NET Core MVC on .NET 8
Database	SQLite database file: GymAkhada.db
Current master UI	Gym Category Master: list page + create form
Main route	/GymCategory and /GymCategory/Create

1. Project Flow

Request flow simple hai: browser URL hit karta hai, routing request ko controller action tak bhejti hai, controller Entity Framework Core se SQLite database me data read/write karta hai, phir Razor view HTML render karta hai.

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Browser -> ASP.NET Core Routing -> GymCategoryController -> ApplicationDbContext -> GymAkhada.db -> Razor View
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2. Folder and File Map

Folder/File	Kaam
.vscode	VS Code workspace/editor settings. App ke runtime logic ka part nahi hota.
bin	.NET build ke compiled DLL/exe output. Isse manually edit nahi karte; build se recreate hota hai.
obj	.NET intermediate build files, generated assets, project cache. Isko bhi manually edit nahi karte.
Controllers	MVC controllers. URL request ko receive karke model/database ke saath kaam karta hai.
Data	Database access layer. EF Core DbContext yahin hai.
Migrations	Database schema history. Tables/columns ka versioned record yahan hota hai.
Models	Entity/data classes. Database table structure aur validation rules yahan define hain.
Properties	Launch profile settings, local run ports, environment variables.
Views	Razor UI files. HTML + C# mix karke browser page render hota hai.
wwwroot	Static files: CSS, JS, Bootstrap, jQuery, favicon. Browser direct serve karta hai.
appsettings.json	Main configuration: connection string, logging, allowed hosts.
appsettings.Development.json	Development environment specific config/logging overrides.
GymAkhada.csproj	.NET project file: framework and NuGet package references.
GymAkhada.db	SQLite database file where records are stored.

3. Important Code Files

File	Kaisa code likha hai	Kya kaam karta hai
Program.cs	Startup/config code: AddControllersWithViews, AddDbContext with SQLite, middleware, default route.	Application start karta hai, MVC enable karta hai, database context register karta hai, static files and routing set karta hai.
Data/ApplicationDbContext.cs	EF Core DbContext code: DbSet, DbSet, OnModelCreating relation.	Models ko database tables se connect karta hai. Category and subcategory ka one-to-many relation define karta hai.
Models/GymCategory.cs	Entity model code: Gym_ID key, Required Gym_categoryName, Gym_Remarks, Subcategories navigation.	Current master form ka data structure define karta hai. GymCategories table isi model se banti hai.
Models/GymSubcategory.cs	Child entity code: GymSub_ID key, Gym_ID foreign key, GymSub_age, GymWeight.	Subcategory/detail data ka structure define karta hai. UI/controller abhi nahi bana, par table/model ready hai.
Controllers/GymCategoryController.cs	MVC action code: Index GET, Create GET, Create POST with anti-forgery and ModelState validation.	List page dikhata hai, form open karta hai, submitted category ko database me save karta hai.
Migrations/InitialCreate.cs	Migration code: CreateTable for GymCategories and GymSubcategories, primary key, foreign key, index.	Database schema banata hai. SQLite file me tables/columns ka structure apply hota hai.

4. View Files and UI

View file	Kaam
Views/GymCategory/Index.cshtml	Gym Category Master list page. Header, Add New Category button, record count, empty state, table.
Views/GymCategory/Create.cshtml	Add Gym Category form. Fields: Category Name, Remarks. Buttons: Save Category, Back to List.
Views/Shared/_Layout.cshtml	Common layout: navbar, footer, Bootstrap/CSS/JS references. Saare pages is layout me render hote hain.
Views/_ViewImports.cshtml	Razor namespaces and tag helper imports. asp-action/asp-controller helpers yahin se available hote hain.
Views/_ViewStart.cshtml	Default layout set karta hai, usually Shared/_Layout.cshtml.
Views/Home/*.cshtml	Default home/privacy pages. Main master flow ka core part nahi.

5. Current Master Form Status

- Haan, current UI ek master form hai: Gym Category Master.
- Visible fields: Category Name (e.g. 15-21) and Remarks.
- Data save target: GymCategories table inside GymAkhada.db.
- GymSubcategory model/table available hai, par uska separate UI/controller abhi nahi hai.
- Master-detail complete karna ho to next step: GymSubcategoryController + Views + category dropdown.

6. Static Assets and Styling

wwwroot/css/site.css me base font sizing, focus styling, dark background, aur newly improved master page classes hain: master-hero, master-panel, master-form, master-help, category-pill. wwwroot/lib me Bootstrap, jQuery, validation scripts already available hain.

7. Database and Migrations

Migrations/20260702075228_InitialCreate.cs database schema create karta hai. Is migration me GymCategories aur GymSubcategories tables, primary keys, foreign key, aur index define hain. GymAkhada.db actual SQLite file hai jisme app runtime data save/read karta hai.

8. Practical Notes

- Controllers me business/request flow samjho.
- Models me database fields aur validation samjho.
- Views me page UI and form binding samjho.
- Data/ApplicationDbContext.cs me tables aur relation samjho.
- Migrations ko manually edit karne se pehle database impact samajhna zaroori hai.
- bin/obj generated folders hain; inme feature code nahi likhte.